ENR 3.2 AREA NAVIGATION ROUTES

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits or (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
L512 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ TENAS 373820N 1313427E							FREQ : 122.250(125.925) MHz 263.350(263.85) MHz
	N/A	099/279	52.3	UNL / FL 270(1 500) Class A, G	↓		Airspace Classification refer to ENR 3.1-1
△ SABET 373829N 1324019E							* L512 OPS HR between TENAS and ANDOL
		094/274	15.7			↑	- EASTBOUND : H24 - WESTBOUND : H24
▲ ANDOL(FIR BDRY) 373958N 1330000E							** After ANDOL, MEA is FL 290, see AIP JAPAN.
INCHEON FIR							*** Extended DME DOC volume service is 220 NM.
FUKUOKA FIR							
1. Critical DME : KAE<TENAS/SABET>, KPO<TENAS/SABET> 2. DME GAP : SABET/ANDOL, GNSS required.							
Y233 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ BUSKO 374033N 1301610E							FREQ : 122.250(125.925) MHz 263.350(263.85) MHz
	N/A	070/250	30.0	UNL / FL 200(1 500) Class A, D, G	↓		Airspace Classification refer to ENR 3.1-1
△ SELPA 375515N 1304911E							
		070/251	69.1				
△ ONATA 382832N 1320602E							
		071/251	20.0			↑	
▲ KANSU(FIR BDRY) 383800N 1322830E							
INCHEON FIR							
PYONGYANG FIR							
1. Critical DME : KAE<BUSKO/SELPA>, KPO<BUSKO/SELPA>, KAE<SELPA/ONATA>, KPO<SELPA/ONATA> 2. DME GAP : ONATA/KANSU, GNSS required.							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y253 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC**
△ DALSU 350731N 1264206E							FREQ: (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz
	N/A	098/278	5.4	UNL / 8 000(2 400) Class A, D, G	↓		FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz
△ GWANGJU VOR(KWA) 350734N 1264844E		098/278	2.6	UNL / 8 000(2 000) Class A, D, G			Airspace Classification refer to ENR 3.1-1
△ SAMUL 350736N 1265154E		098/278	22.1	UNL / 8 000(5 200) Class A, D, G			
▲ TEDAN 350744N 1271852E		098/278	13.5	UNL / 8 000(5 400) Class A, D, G			**Daegu ACC**
△ ANUBA 350746N 1273523E		098/279	44.7				FREQ : 128.175(128.325) MHz 335.50(275.20) MHz
▲ SAPDI 350737N 1282952E		099/279	1.6	UNL / 8 000(2 400) Class A, D, G			Airspace Classification refer to ENR 3.1-1
▲ SARAM 350736N 1283147E		099/279	11.9	UNL / 8 000(4 000) Class A, D, G			Between ANUBA and SAPDI 11 000 ft AMSL to FL 190 VMC-IMC use for training purpose.
△ ANKUS 350730N 1284616E		099/279	11.2	UNL / 8 000(3 500) Class A, D, G			**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
△ BUSAN VORTAC(PSN) 350721N 1285958E					↑		Airspace Classification refer to ENR 3.1-1

1. Critical DME : PSN<SAPDI/SARAM>, CJU<SAPDI/SARAM>, PSN<SARAM/ANKUS>, CJU<SARAM/ANKUS>, PSN<ANKUS/PSN>, CJU<ANKUS/PSN>

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of remarks for Y253.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y437 (RNAV2) [GNSS,DME/DME, DME/DME/IRU]							**Daegu ACC**
△ GANGWON VORTAC(KAE) 374203N 1284514E							FREQ : (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz
	N/A	100/280	44.3	UNL / 8 000(6 400) Class A, D, G	↓		FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz
△ LESBU 374116N 1294104E							Airspace Classification refer to ENR 3.1-1
		100/281	7.8	UNL / 8 000(1 500) Class A, D, G			
▲ UGOVI 374105N 1295051E							**Daegu ACC**
		101/281	20.1				FREQ : 122.250(125.925) MHz 263.350(263.85) MHz
△ BUSKO 374033N 1301610E							Airspace Classification refer to ENR 3.1-1
		101/282	62.2				
△ TENAS 373820N 1313427E							
		044/225	20.0	UNL / FL 200(1 500) Class A, D, G			
△ MALSO 375440N 1314904E							
		045/225	53.3			↑	
▲ KANSU(FIR BDRY) 383800N 1322830E							
INCHEON FIR							
PYONGYANG FIR							

1. Critical DME : KAE<KAE/LESBU>, KPO<KAE/LESBU>, KAE<LESBU/UGOVI>, KPO<LESBU/UGOVI>, KAE<UGOVI/BUSKO>, KPO<UGOVI/BUSKO>, KAE<BUSKO/TENAS>, KPO<BUSKO/TENAS>, KAE<TENAS/MALSO>, KPO<TENAS/MALSO>
2. DME GAP : MALSO/KANSU GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y437.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y571 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
▲ SOSDO 330012N 1262735E							
△ OMKIM 331320N 1264114E	N/A	049/229	17.4	UNL / 11 000(1 500) Class A, D, G	↓		
△ PAPLU 333441N 1270337E		049/229	28.4	UNL / 11 000(2 200) Class A, D, G			
▲ AKPON 334650N 1271953E		056/236	18.2	UNL / 11 000(1 500) Class A, D, G			
▲ NISAV 341519N 1275835E		056/237	42.9	UNL / 11 000(1 800) Class A, D, G			**Daegu ACC** FREQ : 128.175(128.325) MHz 335.50(275.20) MHz Airspace Classification refer to ENR 3.1-1
▲ ANROD 343758N 1282952E		057/237	34.4	UNL / 11 000(2 100) Class A, D, G			
△ POVEM 345523N 1285416E		057/238	26.6	UNL / 11 000(2 800) Class A, D, G			
△ BUSAN VORTAC(PSN) 350721N 1285958E		030/210	12.8	UNL / 11 000(3 000) Class A, D, G			

1. DME GAP : SOSDO/OMKIM, OMKIM/PAPLU, PAPLU/AKPON, AKPON/NISAV GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y571.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y572 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : 128.175(128.325) MHz 335.50(275.20) MHz
△ BUSAN VORTAC(PSN) 350721N 1285958E							
	N/A	249/069	10.1	UNL / 11 000(3 000) Class A, D, G		↓	Airspace Classification refer to ENR 3.1-1
△ OLMUD 350225N 1284916E							
		238/057	21.1	UNL / 11 000(3 200) Class A, D, G			
▲ ENGOT 344834N 1282952E							
		237/057	50.4	UNL / 11 000(2 600) Class A, D, G			
▲ POVOR 341520N 1274400E							
		237/056	26.8	UNL / 11 000(2 000) Class A, D, G			
▲ UPGOS 335733N 1271953E							
		236/056	17.0	UNL / 11 000(1 500) Class A, D, G			**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz
△ BILUM 334613N 1270439E							
		236/056	10.5	UNL / 11 000(1 500) Class A, D, G			Airspace Classification refer to ENR 3.1-1
△ BEPKO 333910N 1265514E							
		231/050	21.9	UNL / 11 000(5 600) Class A, D, G			
△ JEJU VORTAC(CJU) 332305N 1263727E							※ The cruising levels from CJU to RUGMA are even levels due to operational reasons.
		170/350	10.2	UNL / 11 000(8 700) Class A, D, G		↓	
△ OMKIM 331320N 1264114E							
		170/350	13.8	UNL / 11 000(1 500) Class A, D, G			※ The cruising levels from RUGMA to CJU are odd levels due to operational reasons.
▲ TOSAN 330012N 1264619E							
		170/349	31.5	UNL / 11 000(1 500) Class A, D, G			
▲ RUGMA(FIR BDRY) 323012N 1265753E					↑		
INCHEON FIR							
FUKUOKA FIR							

1. DME GAP : UPGOS/BILUM, BILUM/BEPKO, BEPKO/CJU, CJU/OMKIM, OMKIM/TOSAN, TOSAN/RUGMA GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y572.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y579 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
△ TENAS 373820N 1313427E							
△ AGSUS 364521N 1304044E	N/A	228/048	68.1	UNL / FL 140(1 500) Class A, D, G			
▲ DABIK 361743N 1301143E		229/049	36.2				
▲ BULGA 355609N 1294924E		229/049	28.1				**Daegu ACC** FREQ : 120.575(119.375, 119.325, 134.375) MHz 254.70(335.75) MHz Airspace Classification refer to ENR 3.1-1
▲ BEDOM 352513N 1291754E		229/048	40.2	UNL / FL 140(2 700) Class A, D, G			
△ BUSAN VORTAC(PSN) 350721N 1285958E		228/048	23.1	UNL / FL 140(3 800) Class A, D, G			

1. TENAS-PSN CDR1 Operational hour(UTC) - Weekdays : 1400~2200 - SAT: 2200 on the preceding until 2400 on the Saturday - SUN : 0000~2200 - Holiday : 1400 on the preceding until 2200 on the holiday. Rest of Y579-PERM. See ENR 1.1-1.2.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y579.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y644 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Eastbound (AGAVO-EGOBA) FL 410, FL 390, FL 370, FL 350, FL 330, FL 310, FL 290, FL 270, FL 250, FL 230 REF. ENR 3.1-10 for the detailed altitude conversion procedures. Airspace Classification refer to ENR 3.1-1
SANGHAI FIR							
INCHEON FIR							
▲ AGAVO(FIR BDRY) 371000N 1240000E							
△ RILRO 371033N 1241442E	N/A	096/276	11.8	UNL / FL 150(1 500) Class A, D, G	↓		
△ GONAV 371048N 1242453E		097/277	8.1				
△ BODOL 371122N 1244954E		097/277	20.0				
▲ REBIT 371203N 1252913E		097/278	31.4				
△ BELTU 371218N 1254759E		098/278	15.0				
△ BOGAN 371241N 1262812E		098/278	32.1	UNL / 8 000(2 400) Class A, D, G			
▲ MONSI 371247N 1265015E		098/279	17.6	UNL / 8 000(3 200) Class A, D, G			
△ POLEG 371249N 1265935E		099/279	7.5	UNL / 8 000(2 400) Class A, D, G			
△ EGOBA 372915N 1272246E		057/237	24.7	UNL / FL 140(3 300) Class A, D, G			

1. Critical DME : SEL<AGAVO/RILRO>, KUZ<AGAVO/RILRO>, SEL<RILRO/GONAV>, KUZ<RILRO/GONAV>, SEL<MONSI/POLEG>, SOT<MONSI/POLEG>, SEL<POLEG/EGOBA>, SOT<POLEG/EGOBA>

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y644.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y655 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Airspace Classification refer to ENR 3.1-1
△ GONAV 371048N 1242453E							
△ DALPO 365835N 1242453E	N/A	189/008	12.2	UNL / FL 140(1 500) Class A, D, G	↓		
▲ NONOS 364046N 1242453E		188/008	17.8				
▲ BIDRI 362007N 1242453E		188/008	20.6				
▲ DANPA 353036N 1242453E		188/008	49.4				**Incheon ACC** FREQ : (below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz FREQ : (at or above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz Airspace Classification refer to ENR 3.1-1
▲ PALSA 340131N 1242453E		188/008	88.9				
△ TOLIS 335030N 1242453E		188/008	11.0				**Incheon ACC** FREQ : (below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (at or above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
▲ ENSUM 321302N 1244635E		177/357	99.0	UNL / FL 430(1 500) Class A, G			**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
△ BONSO 302840N 1250851E		177/356	105.9	FL 220 / FL 150(1 500) Class A, D			**Incheon ACC** FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz
▲ ATOTI(FIR BDRY) 300013N 1251154E		181/001	28.5			↑	FREQ : (at or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz
INCHEON FIR							Airspace Classification refer to ENR 3.1-1
FUKUOKA FIR							

1. Critical DME : SEL<GONAV/DALPO>, KUZ<GONAV/DALPO>, SEL<DALPO/NONOS>, KUZ<DALPO/NONOS>, SEL<NONOS/BIDRI>, KUZ<NONOS/BIDRI>, SEL<BIDRI/DANPA>, KUZ<BIDRI/DANPA>
2. DME GAP : DANPA/PALSA, PALSA/TOLIS, TOLIS/ENSUM, ENSUM/BONSO, BONSO/ATOTI GNSS required.

Y657 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC** FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
△ GWANGJU VOR(KWA) 350734N 1264844E	N/A	073/253	54.7	10 000 / 8 000(7 100) Class D	↓		
▲ IGDOK 353104N 1274907E		073/254	41.6	10 000 / 8 000(4 800) Class D		↑	**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
△ DALSEONG VORTAC(TGU) 354835N 1283527E							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y655 and Y657.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y659 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC** FREQ: (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz FREQ : (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz Airspace Classification refer to ENR 3.1-1
△ GUNSAN VORTAC(KUZ) 355437N 1263641E							
△ ELPOS 355410N 1264707E	N/A	101/282	8.5	10 000 / 7 000(1 700) Class D	↓		
△ RINBO 355352N 1265349E		102/282	5.5				
△ MELES 355251N 1271542E		102/282	17.8	10 000 / 7 000(3 700) Class D			
△ OPEDA 355149N 1273652E		102/282	17.2	10 000 / 7 000(5 000) Class D			
△ DALSEONG VORTAC(TGU) 354835N 1283527E		102/283	47.7	10 000 / 7 000(6 600) Class D			**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
▲ LAPAL 355413N 1290452E		085/266	24.6	UNL / 6 000(4 200) Class A, D, G			
△ POHANG VORTAC(KPO) 355838N 1292828E		086/266	19.7	UNL / 6 000(3 300) Class A, D, G		↑	
Y677 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
△ TOLIS 335030N 1242453E							
▲ LIMDI 333313N 1254953E	N/A	111/292	73.0	UNL / 9 000(1 500) Class A, D, G	↓		
△ REMOS 332605N 1262329E		112/292	29.0	UNL / 9 000(4 100) Class A, D, G			
△ JEJU VORTAC(CJU) 332305N 1263727E		112/292	12.1	UNL / 9 000(8 700) Class A, D, G			
▲ TAMNA 332815N 1271953E		089/270	35.9	UNL / 9 000(6 300) Class A, D, G			
▲ SAMDO(FIR BDRY) 333503N 1281857E		090/270	49.8	UNL / 9 000(1 500) Class A, D, G		↑	
INCHEON FIR							
FUKUOKA FIR							

1. Critical DME : KWA<LIMDI/REMOS>, CJU<LIMDI/REMOS>, PSN<CJU/TAMNA>, CJU<CJU/TAMNA>, PSN<TAMNA/SAMDO>, CJU<TAMNA/SAMDO>
2. DME GAP : TOLIS/LIMDI, REMOS/CJU GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y659 and Y677.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y685 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ ANYANG VORTAC(SEL) 372449N 1265542E							FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
△ KALMA 371845N 1270645E	N/A	133/313	10.7	UNL / 8 000(3 200) / Class A, D, G	↓		Only flying westbound from KPO to SEL on Y685 is authorized except ACFT departing from RKTY or RKTI.
△ KAKSO 370745N 1272637E		133/314	19.3	UNL / 8 000(2 900) / Class A, D, G			Aircraft flying eastbound from SEL to KPO at or above 11 000 ft on Y685 shall get PPR 24 hours before from Incheon/Daegu ACC.
▲ GUKDO 370111N 1273823E		134/314	11.5	UNL / 8 000(3 600) / Class A, D, G			No PPR is needed at or below 10 000 ft. Airspace Classification refer to ENR 3.1-1
△ ENSAL 365554N 1274747E		134/314	9.2	UNL / 8 000(3 700) / Class A, D, G			**Daegu ACC**
△ BASEM 365037N 1275710E		134/314	9.2	UNL / 8 000(4 000) / Class A, D, G			FREQ : 120.575(119.375, 119.325, 134.375) MHz 254.70(335.75) MHz
▲ BIGOB 364325N 1280952E		134/314	12.5	UNL / 8 000(5 000) / Class A, D, G			Only flying westbound from KPO to SEL on Y685 is authorized except ACFT departing from RKTY or RKTI.
△ YECHEON VOR(CUN) 363755N 1281931E		134/314	9.5	UNL / 8 000(4 900) / Class A, D, G			Aircraft flying eastbound from SEL to KPO at or above 11 000 ft on Y685 shall get PPR 24 hours before from Incheon/Daegu ACC.
△ ELAPI 362014N 1285051E		134/314	30.8	UNL / 8 000(3 800) / Class A, D, G			No PPR is needed at or below 10 000 ft.
△ POHANG VORTAC(KPO) 355838N 1292828E		134/314	37.3	UNL / 8 000(4 700) / Class A, D, G			Airspace Classification refer to ENR 3.1-1
▲ BULGA 355609N 1294924E		107/287	17.2	UNL / 8 000(2 100) / Class A, D, G			
▲ SAPRA(FIR BDRY) 354926N 1304325E		107/288	44.4	UNL / 8 000(1 500) / Class A, D, G		↑	
INCHEON FIR							
FUKUOKA FIR							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y685.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y697 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
SANGHAI FIR							
INCHEON FIR							
▲ AGAVO(FIR BDRY) 371000N 1240000E							FREQ: (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
▲ OLBIM 371411N 1240751E	N/A	065/245	7.5	UNL / FL 150(1 500) Class A, D, G			Westbound(SEL-AGAVO) FL 400, FL 380, FL 360, FL 340, FL 320, FL 300, FL 280, FL 260, FL 240, FL 220, FL 200
△ NOGON 372250N 1242505E		066/247	16.2				REF. ENR 3.1-9 for the detailed altitude conversion procedures.
△ ANSIM 372323N 1245009E		097/277	20.0				Only flying westbound from SEL to AGAVO on Y697 is authorized.
▲ BINIL 372349N 1251359E		097/277	19.0				
▲ NOPIK 372412N 1253905E		097/278	20.0				
△ GOGET 372442N 1263036E		098/278	41.0	UNL / 8 000(2 100) Class A, D, G			
△ ANYANG VORTAC(SEL) 372449N 1265542E		098/279	20.0	UNL / 7 500(3 400) Class A, D, G	↓	↑	
△ EGOBA 372915N 1272246E		087/267	22.0				
▲ KARBU 373159N 1273952E		088/268	13.9	UNL / 7 500(5 100) Class A, D, G			
▲ TORUS 373625N 1280807E		088/268	22.9	UNL / 7 500(4 500) Class A, D, G			**Daegu ACC** FREQ: (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz
▲ BIKSI 374032N 1283504E		088/268	21.8	UNL / 7 500(7 500) Class A, D, G			FREQ: (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz
△ GANGWON VORTAC(KAE) 374203N 1284514E		088/269	8.2	UNL / 7 500(7 100) Class A, D, G			Airspace Classification refer to ENR 3.1-1
▲ PILIT 372631N 1291731E		130/310	30.0	UNL / 8 000(7 100) Class A, D, G			
▲ ESNEG 371014N 1295051E		130/311	31.2	UNL / 9 000(1 500) Class A, D, G			
△ AGSUS 364521N 1304044E		131/311	47.1				**Daegu ACC** FREQ: 122.250(125.925) MHz 263.350(263.85) MHz Only flying westbound from LANAT to KAE on Y697 shall get 24HR PPR from Daegu ACC.
▲ LANAT(FIR BDRY) 362224N 1312542E		131/311	42.9			↑	Airspace Classification refer to ENR 3.1-1
INCHEON FIR							
FUKUOKA FIR							

1. Critical DME : SEL<AGAVO/OLBIM>, KUZ<AGAVO/OLBIM>, SEL<OLBIM/NOGON>, KUZ<OLBIM/NOGON>, SEL<NOGON/ANSIM>, KUZ<NOGON/ANSIM>, SEL<KARBU/TORUS>, KAE<KARBU/TORUS>, SEL<TORUS/BIKSI>, KAE<TORUS/BIKSI>, SEL<BIKSI/KAE>, KAE<BIKSI/KAE>, KAE<KAE/PILIT>, KPO<KAE/PILIT>, KAE<PILIT/ESNEG>, KPO<PILIT/ESNEG>, KAE<ESNEG/AGSUS>, KPO<ESNEG/AGSUS>, KAE<AGSUS/LANAT>, KPO<AGSUS/LANAT>

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y697.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels: Odd	Direction of cruising levels: Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y711 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Airspace Classification refer to ENR 3.1-1
▲ MONSI 371247N 1265015E							
▲ BULTI 364322N 1264930E	N/A	190/010	29.4	UNL / FL 140(2 900) Class A, D, G			
▲ MEKIL 363322N 1264953E		187/007	10.0				**Incheon ACC** FREQ : (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz FREQ : (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz Airspace Classification refer to ENR 3.1-1
△ GONAX 362311N 1265016E		187/007	10.2	UNL / FL 140(3 200) Class A, D, G			
△ BEDES 360905N 1264844E		194/014	14.1				
△ ELPOS 355410N 1264707E		194/013	14.9	UNL / FL 140(2 200) Class A, D, G			
▲ MANGI 353011N 1264432E		193/013	24.0	UNL / FL 140(2 800) Class A, D, G			
△ DALSU 350731N 1264206E		193/013	22.7	UNL / FL 140(3 800) Class A, D, G			**Incheon ACC** FREQ : (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz Airspace Classification refer to ENR 3.1-1
△ NULDI 342514N 1263739E		193/013	42.4	UNL / FL 140(4 000) Class A, D, G			
▲ DOTOL 341515N 1263637E		193/013	10.0	UNL / FL 140(3 300) Class A, D, G			
△ KIDOS 335028N 1263402E		193/013	24.8	UNL / FL 140(2 700) Class A, D, G			**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
△ REMOS 332605N 1262329E		208/028	25.9	UNL / FL 140(6 000) Class A, D, G			
▲ PANSI 330014N 1261225E		208/027	27.4	UNL / FL 140(6 300) Class A, D, G			
△ DOMKO 322848N 1255859E		207/027	33.4	UNL / FL 140(1 500) Class A, D, G			**Incheon ACC** FREQ : (At or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz * Note Any aircraft planning to operate below FL 320 after MUGUS must use Z86 between BONSO and ATOTI, then use B576. Airspace Classification refer to ENR 3.1-1
△ PONIK 320021N 1254659E		207/027	30.1				
△ IKEDO 314314N 1253948E		207/027	18.2				
▲ KANKA 313155N 1253504E		207/027	12.0				
△ BONSO 302840N 1250851E		207/026	67.0				
▲ MUGUS(FIR BDRY) 300006N 1245712E		206/026	30.2				
INCHEON FIR							
FUKUOKA FIR							

1. Critical DME : KWA<DOTOL/KIDOS>, CJU<DOTOL/KIDOS>
2. DME GAP : KIDOS/REMOS, REMOS/PANSI, PANSI/DOMKO, DOMKO/PONIK, PONIK/IKEDO, IKEDO/KANKA, KANKA/BONSO, BONSO/MUGUS GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track and DME GAP for Y711.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels: Odd	Direction of cruising levels: Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y722 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC**
△ SONGTAN VORTAC(SOT) 370540N 1270154E							FREQ: (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz
▲ OLMEN 364413N 1265928E	N/A	194/014	21.5	UNL / FL 140(3 100) / Class A, D, G			FREQ : (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz
△ GUNKU 363414N 1265949E		187/007	10.0	UNL / FL 140(3 600) / Class A, D, G			Airspace Classification refer to ENR 3.1-1
▲ PEBRI 362311N 1270013E		187/007	11.0	UNL / FL 140(3 300) / Class A, D, G			
△ ATASO 355344N 1265657E		194/014	29.5	UNL / FL 140(2 300) / Class A, D, G			
▲ MAKSA 353011N 1265422E		194/013	23.6	UNL / FL 140(3 800) / Class A, D, G			**Incheon ACC**
△ SAMUL 350736N 1265154E		193/013	22.7	UNL / FL 140(4 000) / Class A, D, G			FREQ: (at or below FL 255) 120.725(128.30) MHz 263.90(272.75) MHz
▲ KAMIT 341514N 1264618E		193/013	52.5	UNL / FL 140(3 400) / Class A, D, G			FREQ : (above FL 255) 123.725(124.50) MHz 239.25(275.40) MHz
△ GUKSU 335251N 1264357E		193/013	22.4	UNL / FL 140(2 100) / Class A, D, G			Airspace Classification refer to ENR 3.1-1
△ LOSNI 333315N 1264153E		193/013	19.6	UNL / FL 140(1 700) / Class A, D, G			**Incheon ACC**
△ JEJU VORTAC(CJU) 332305N 1263727E		208/028	10.8	UNL / FL 140(7 600) / Class A, D, G			FREQ: 124.525(132.425) MHz 255.40(233.50, 348.10) MHz
▲ SOSDO 330012N 1262735E		208/027	24.3	UNL / FL 140(8 700) / Class A, D, G			Airspace Classification refer to ENR 3.1-1
△ SAMLO 323223N 1261536E		208/027	29.5	UNL / FL 140(1 500) / Class A, D, G			
△ NIRAT 320354N 1260329E		207/027	30.2				**Incheon ACC**
△ ELGEP 314653N 1255617E		207/027	18.1				FREQ: (At or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz
▲ TESIM 313526N 1255128E		207/027	12.1				FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz
▲ ATOTI(FIR BDRY) 300013N 1251154E		207/026	100.9		↑		Airspace Classification refer to ENR 3.1-1
INCHEON FIR							
SANGHAI FIR							

1. Critical DME : KWA<KAMIT/GUKSU>, CJU<KAMIT/GUKSU>
2. DME GAP : GUKSU/LOSNI, LOSNI/CJU, CJU/SOSDO, SOSDO/SAMLO, SAMLO/NIRAT, NIRAT/ELGAP, ELGAP/TESIM, TESIM/ATOTI, GNSS required.

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track and DME GAP for Y722.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y744 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz 1. 11 000 ft to FL 240, at or above FL 280 will be blocked. 2. At or above 11 000 ft, required 15 days PPR from Air Traffic Management Office. 3. Airspace Classification refer to ENR 3.1-1
▲ PILIT 372631N 1291731E	N/A	183/003	19.3	UNL / 9 000(4 900) Class A, D, G			
▲ NOBUT 370715N 1291957E		183/003	28.6	UNL / 9 000(5 600) Class A, D, G			
▲ OSVOM 363844N 1292331E		183/003	18.5	UNL / 9 000(4 000) Class A, D, G			**Daegu ACC** FREQ : 120.575(119.375, 119.325, 134.375) MHz 254.70(335.75) MHz 1. 11 000 ft to FL 240, at or above FL 280 will be blocked. 2. At or above 11 000 ft, required 15 days PPR from Air Traffic Management Office. 3. Airspace Classification refer to ENR 3.1-1
▲ LOSTO 362016N 1292548E		183/003	21.7	UNL / 9 000(2 100) Class A, D, G			
△ POHANG VORTAC(KPO) 355838N 1292828E		213/033	37.2	UNL / 9 000(4 400) Class A, D, G			
▲ APARU 352442N 1290932E		213/033	19.0	UNL / 9 000(4 400) Class A, D, G			
△ BUSAN VORTAC(PSN) 350721N 1285958E							
1. Critical DME : KAE<PILIT/NOBUT>, KPO<PILIT/NOBUT>							
Y781 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
△ DALSEONG VORTAC(TGU) 354835N 1283527E	N/A	193/013	19.8	UNL / 7 000(4 900) Class A, D, G			
▲ MASTA 352847N 1283340E		162/343	23.6	UNL / 7 000(4 000) Class A, D, G			
△ ANKUS 350730N 1284616E		163/343	7.7	UNL / 7 000(3 500) Class A, D, G			
△ OMOTU 350033N 1285022E		156/336	27.3	UNL / 7 000(2 800) Class A, D, G			
▲ BESNA(FIR BDRY) 343718N 1290751E							
INCHEON FIR							
FUKUOKA FIR							
1. Critical DME : PSN<OMOTU/BESNA>, CJU<OMOTU/BESNA> 2. DME GAP : ANKUS/OMOTU, GNSS required.							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y744 and Y781.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y782 (RNAV2) [GNSS, DME/DME, DME/DME/IRU] △ ANYANG VORTAC(SEL) 372449N 1265542E △ POLEG 371249N 1265935E △ SONGTAN VORTAC(SOT) 370540N 1270154E ▲ OSPOT 365018N 1272055E ▲ VASLI 364252N 1273003E ▲ MAKDU 362712N 1274909E ▲ BITUX 361645N 1280148E △ DALSEONG VORTAC(TGU) 354835N 1283527E ▲ KALOD 353012N 1284626E △ BUSAN VORTAC(PSN) 350721N 1285958E ▲ APELA(FIR BDRY) 344323N 1291400E							**Daegu ACC** FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Airspace Classification refer to ENR 3.1-1
	N/A	174/354	12.4	UNL / 4 500(3 000) Class A, D, G	↓		
		174/354	7.4	UNL / 4 500(2 000) Class A, D, G			
		144/324	21.6	UNL / 8 000(3 300) Class A, D, G			
		144/324	10.4				**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz Airspace Classification refer to ENR 3.1-1
		144/324	21.9	UNL / 8 000(4 200) Class A, D, G			
		144/324	14.6				
		144/325	39.2	UNL / 10 000(4 500) Class A, D, G			
		162/343	20.4	UNL / 8 000(4 400) Class A, D, G			
		162/343	25.4	UNL / 5 000(3 600) Class A, D, G			
		163/343	26.6	UNL / 4 000(3 000) Class A, D, G		↑	
INCHEON FIR							
FUKUOKA FIR							

1. Critical DME : SEL<SEL/POLEG>, SOT<SEL/POLEG>, SEL<POLEG/SOT>, SOT<POLEG/SOT>

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y782.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Z50 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ EGOBA 372915N 1272246E							FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
	N/A	224/044	28.8	UNL / FL 140(3 300) Class A, D, G		↓	FREQ: (above FL295) 132.80(120.525) MHz 290.60(335.45) MHz
△ SONGTAN VORTAC(SOT) 370540N 1270154E		213/033	24.4	UNL / FL 140(2 900) Class A, D, G	↑		Airspace Classification refer to ENR 3.1-1
▲ BULTI 364322N 1264930E							
Z51 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC**
▲ BOPTA 364406N 1263658E							FREQ: (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz
	N/A	173/353	21.6	UNL / FL 150(3 900) Class A, D, G		↓	FREQ: (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz
△ MOXID 362311N 1264359E		173/353	14.6	UNL / FL 150(3 600) Class A, D, G			Airspace Classification refer to ENR 3.1-1
△ BEDES 360905N 1264844E							
Z52 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Incheon ACC**
▲ OLMEN 364413N 1265928E							FREQ: (at or below FL 255) 126.175(134.375) MHz 317.85(335.55) MHz
	N/A	051/232	16.4	UNL / 8 000(3 400) Class A, D, G	↓		FREQ: (above FL 255) 132.15(123.55) MHz 263.15(272.60) MHz
△ POSAN 365615N 1271316E		052/232	15.7	UNL / 8 000(3 000) Class A, D, G		↑	Airspace Classification refer to ENR 3.1-1
△ KAKSO 370745N 1272637E							
Z53 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
▲ BITUX 361645N 1280148E							FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
	N/A	002/182	17.0	UNL / FL 160(4 000) Class A, D, G		↓	FREQ: (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
△ TEBEX 363341N 1275929E		002/182	17.0	UNL / FL 160(4 600) Class A, D, G	↑		Airspace Classification refer to ENR 3.1-1
△ BASEM 365037N 1275710E							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Z50 and Z52.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Z54 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ SONGTAN VORTAC(SOT) 370540N 1270154E							FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
▲ MONSI 371247N 1265015E	N/A	316/136	11.7	UNL / 8 000(2 400) Class A, D, G		↓	FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
△ GOGET 372442N 1263036E		316/136	19.7	UNL / 8 000(2 300) Class A, D, G	↑		Airspace Classification refer to ENR 3.1-1
Z55 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
SANGHAI FIR							FREQ: (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz
INCHEON FIR							
▲ AGAVO(FIR BDRY) 371000N 1240000E							FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz
▲ NONOS 364046N 1242453E	N/A	154/334	35.4	UNL / FL 140(1 500) Class A, D, G		↑	Airspace Classification refer to ENR 3.1-1
1. Critical DME : SEL<AGAVO/NONOS>, KUZ<AGAVO/NONOS>							
Z56 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
PYONGYANG FIR							FREQ : 122.250(125.925) MHz 263.350(263.85) MHz
INCHEON FIR							
▲ KANSU(FIR BDRY) 383800N 1322830E							Airspace Classification refer to ENR 3.1-1
△ PALDU 375813N 1323625E	N/A	180/360	40.2	UNL / FL 200(1 500) Class A, G	↓		
△ SABET 373829N 1324019E		180/360	19.9				
▲ IGRAS(FIR BDRY) 371846N 1324411E		180/360	19.9			↑	
INCHEON FIR							
FUKUOKA FIR							
1. Critical DME : KAE<PALDU/SABET>, KPO<PALDU/SABET>, KAE<SABET/IGRAS>, KPO<SABET/IGRAS> 2. DME GAP : KANSU/PALDU, GNSS required.							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Z57 **(RNAV2)** **[GNSS, DME/DME, DME/DME/IRU]** △ RILRO 371033N 1241442E △ DALPO 365835N 1242453E	N/A	154 / 334	14.5	UNL / FL 150(1 500) Class A, D, G	↓		**Daegu ACC** FREQ : (at or below FL 295) 128.70(118.925) MHz 270.50(263.60) MHz FREQ : (above FL 295) 132.80(120.525) MHz 290.60(335.45) MHz Airspace Classification refer to ENR 3.1-1
1. Critical DME : SEL<RILRO/DALPO>, KUZ<RILRO/DALPO>							
Z63 **(RNAV2)** **[GNSS, DME/DME, DME/DME/IRU]** ▲ PILIT 372631N 1291731E △ LESBU 374116N 1294104E	N/A	061 / 241	23.8	UNL / FL 250(1 500) Class A, G	↓	↑	**Daegu ACC** FREQ : (at or below FL 295) 134.175(123.65) MHz 272.40(233.60) MHz FREQ : (above FL 295) 122.250(125.925) MHz 263.350(263.85) MHz Airspace Classification refer to ENR 3.1-1
1. Critical DME : KAE<PILIT/LESBU>, KPO<PILIT/LESBU>							
Z81 **(RNAV2)** **[GNSS, DME/DME, DME/DME/IRU]** △ KIDOS 335028N 1263402E △ JEJU VORTAC(CJU) 332305N 1263727E	N/A	182 / 002	27.5	UNL / FL 140(8 700) Class A, D, G		↓	**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
Z82 **(RNAV2)** **[GNSS, DME/DME/IRU]** △ JEJU VORTAC(CJU) 332305N 1263727E ▲ PANSI 330014N 1261225E	N/A	230 / 050	31.0	UNL / FL 140(8 700) Class A, D, G		↓	**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz Airspace Classification refer to ENR 3.1-1
1. DME GAP : CJU/PANSI, GNSS required.							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Z63 and Z81.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Z83 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ DALSEONG VORTAC(TGU) 354835N 1283527E							FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
▲ MASTA 352847N 1283340E	N/A	193/013	19.8	UNL / 5 000(4 900) Class A, D, G		↓	Airspace Classification refer to ENR 3.1-1
▲ SARAM 350736N 1283147E		193/013	21.2	UNL / 5 000(3 800) Class A, D, G			
▲ ENGOT 344834N 1282952E		193/013	19.1	UNL / 5 000(3 000) Class A, D, G			
▲ ANROD 343758N 1282952E		188/008	10.6	UNL / 5 000(2 300) Class A, D, G	↑		
1. Critical DME : PSN<TGU/MASTA>, TGU<TGU/MASTA>, PSN<MASTA/SARAM>, TGU<MASTA/SARAM>, PSN<SARAM/ENGOT>, TGU<SARAM/ENGOT> 2. DME GAP : ENGOT/ANROD, GNSS required.							
Z84 (RNAV2) [GNSS, DME/DME, DME/DME/IRU]							**Daegu ACC**
△ BUSAN VORTAC(PSN) 350721N 1285958E							FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
▲ KALEK(FIR BDRY) 351232N 1295305E	N/A	091/272	43.8	UNL / 8 000(3 100) Class A, D, G	↓	↑	Airspace Classification refer to ENR 3.1-1
INCHEON FIR							
FUKUOKA FIR							
Z85 (RNAV2) [GNSS, DME/DME/IRU]							**Incheon ACC**
△ BILUM 334613N 1270439E							FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz
△ PAPLU 333441N 1270337E	N/A	192/012	11.5	UNL / FL 170(1 500) Class A, D, G		↓	Airspace Classification refer to ENR 3.1-1
▲ RUGMA(FIR BDRY) 323012N 1265753E		192/012	64.5		↑		
INCHEON FIR							
FUKUOKA FIR							
1. DME GAP : BILUM/PAPLU, PAPLU/RUGMA, GNSS required.							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Z83 and Z85.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Z86 **(RNAV2)** **[GNSS, DME/DME/IRU]** △ BONSO 302840N 1250851E							**Incheon ACC** FREQ : 124.525(132.425) MHz 255.40(233.50, 348.10) MHz
	N/A	181 / 001	28.5	UNL / FL 140(1 500) Class A, D, G		↓	* Note Any aircraft planning to operate below FL 320 after MUGUS must use Z86 between BONSO and ATOTI, then use B576.
▲ ATOTI(FIR BDRY) 300013N 1251154E							
INCHEON FIR							Airspace Classification refer to ENR 3.1-1
FUKUOKA FIR							
1. DME GAP : BONSO/ATOTI, GNSS required.							
Z91 **(RNAV2)** **[GNSS, DME/DME, DME/DME/IRU]** △ BUSAN VORTAC(PSN) 350721N 1285958E							**Daegu ACC** FREQ : 125.375(125.775, 124.575) MHz 234.15(317.35, 335.50) MHz
	N/A	150 / 330	25.6	UNL / 10 000(3 000) Class A, D, G	↓	↑	Airspace Classification refer to ENR 3.1-1
▲ INVOK(FIR BDRY) 344719N 1291923E							
INCHEON FIR							
FUKUOKA FIR							

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Z91.

Route designator (Navigation specification) Name of significant points Coordinates [Navigation Specification limitation]	Waypoint IDENT of VOR/DME BRG & DIST ELEV DME Antenna	MAG TRACK °	Geodetic DIST NM	Upper limits / Lower limits (MOCA) ft AMSL or FL / Airspace classification	Direction of cruising levels Odd	Direction of cruising levels Even	Remarks Controlling unit Frequency
1	2	3	4	5	6		7
Y590 (RNAV2) [GNSS, DME/DME/IRU]							**Incheon ACC**
FUKUOKA FIR							FREQ : (at or above FL 335) 133.425(132.425) MHz 234.35(234.65) MHz
INCHEON FIR							
▲ BEDAR(FIR BDRY) 315401N 1262910E							FREQ : (below FL 335) 125.725(132.825, 128.375) MHz 232.95(233.15) MHz
	N/A	263 / 083	28.9	UNL / FL 240(1 500) Class A, G			Airspace Classification refer to ENR 3.1-1
△ ELGEP 314653N 1255617E							
		263 / 083	14.5				
△ IKEDO 314314N 1253948E							
		288 / 108	34.5		↑		
▲ SADLI 314948N 1250000E							

1. DME GAP : BEDAR/SADLI, GNSS required.
2. Flight Level Allocation Scheme (FLAS)
   - For the eastbound over BEDAR : FL 250, FL 290, FL 310, FL 390

* RNAV2 represents a navigation accuracy of ± 2 NM on a 95% containment basis.

Change : Information of MAG track for Y590.

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